

Reliability Assessment of the JAXA H3 Fairing under Manufacturing Variability

Polynomial Chaos Expansion as a Stochastic FEM Surrogate

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Stochastic Finite Element Methods — Dr. Z. Zheng / Prof. U. Nackenhorst LUH SoSe 2026

Motivation: manufacturing variability in composite fairings

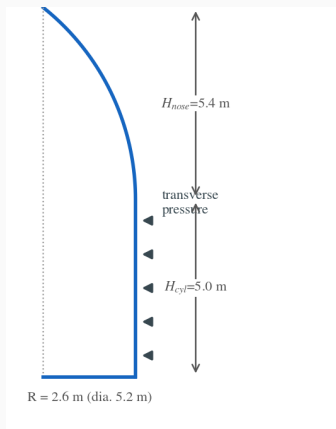
- The H3 fairing is a **CFRP / aluminum-honeycomb sandwich**.
- Material properties vary from manufacturing: fiber volume fraction, cure cycle, adhesive film thickness.
- This uncertainty propagates into **stress, damage, and deflection** — and hence reliability.

Goal: quantify how five uncertain material parameters propagate to the structural response, treating the Abaqus model as a black box.

Stochastic FEM

Propagate input uncertainty through a deterministic FE model to obtain response statistics and a failure probability.

The structure: H3 payload fairing — design & FE model



Geometry (CFRP / Al-honeycomb sandwich)

- Diameter 5.2 m; ogive nose $H_{nose} = 5.4\text{ m}$, cylinder $H_{cyl} = 5.0\text{ m}$
- CFRP face sheets 0.6–1.4 mm; Al-honeycomb core 8 mm

Material constants

- CFRP skin (orthotropic LAMINA): $E_1 \approx 146\text{ GPa}$, $E_2 \approx 10\text{ GPa}$, $G_{12} \approx 5.2\text{ GPa}$, $\nu_{12} = 0.3$
- Al-honeycomb core: orthotropic — stiff out-of-plane (E_3), soft in-plane
- Adhesive — cohesive zone (BK, $\eta = 2.284$): $K_n = 10^6\text{ N/mm}^3$, $G_{IC} = 0.3\text{ N/mm}$, $t_n = 50\text{ MPa}$

Boundary conditions & load

- Base ring **clamped**; skins–core tied via the cohesive interface
- Step 1: transverse **pressure**; Step 2: incremental load

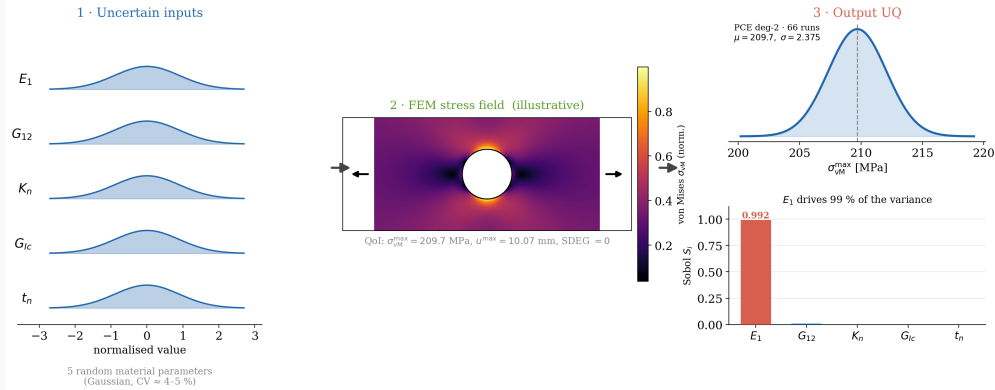
Five constants (E_1 , G_{12} , K_n , G_{IC} , t_n) are treated as **uncertain** (next slides).

Ogive nose over a cylindrical barrel. The UQ study takes a

representative wall panel of this shell.

Approach at a glance

Stochastic FEM: Uncertainty Propagation through a CFRP Composite Panel



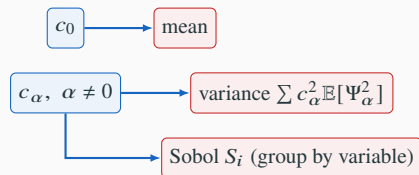
Non-intrusive: sample the five parameters, propagate each draw through the deterministic Abaqus model, recover the response distribution and Sobol indices.

Method: Polynomial Chaos Expansion

For a response $Y = \mathcal{M}(\xi)$ with random inputs $\xi \in \mathbb{R}^d$:

$$Y \approx \hat{Y} = \sum_{\alpha \in \mathcal{A}} c_{\alpha} \Psi_{\alpha}(\xi), \quad P = \binom{d+p}{p}.$$

- Ψ_{α} : **Hermite** polynomials — orthogonal w.r.t. the Gaussian measure.
- The basis is *fixed by the input measure*, not chosen.
- It is the **orthogonality**, not the polynomial form, that pays.



No extra model runs — all three come from the coefficients.

Method: Smolyak sparse quadrature

Full tensor-product quadrature needs n_q^d points — prohibitive at $d = 5$. Smolyak combines low-order 1-D rules, keeping exactness on the **total-degree** space only.



full tensor: 49



Smolyak: 17

(2-D illustration — the same idea at $d = 5$.)

p	Full	Sparse	Reduction
2	$3^5=243$	66	3.7×
3	$4^5=1024$	286	3.6×

Five uncertain input parameters

Parameter	Description	Nominal	CoV
E_1	fiber-direction modulus	146,000 MPa	5%
G_{12}	in-plane shear modulus	5,200 MPa	10%
K_n	cohesive normal stiffness	10^6 N/mm ³	15%
G_{Ic}	Mode-I fracture energy	0.3 N/mm	20%
t_n	cohesive strength	50 MPa	15%

Independent **truncated-normal** variables on $[0.5\mu, 1.5\mu]$ — stiffnesses and fracture energies are strictly positive, and flight hardware is screened against inspection limits.

In σ -units the bound is $0.5/\text{CoV}$, so the truncation is **practically inactive** for every input — it guarantees positivity without reshaping the distribution.

CoV	bound	mass cut
5 % (E_1)	$\pm 10\sigma$	~ 0
15 %	$\pm 3.3\sigma$	0.09 %
20 % (G_{Ic})	$\pm 2.5\sigma$	1.2 %

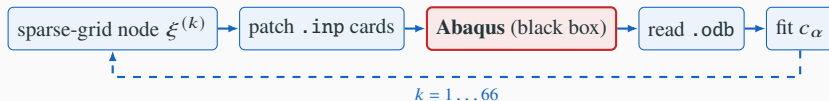
Finite-element model & quantities of interest

Abaqus/Standard panel model:

- CFRP skin: orthotropic LAMINA elastic.
- Al-honeycomb core: linear elastic.
- Adhesive: **cohesive zone model** (BK mixed-mode, $\eta = 2.284$).
- Two-step static load; QoIs from last frame via odbAccess.

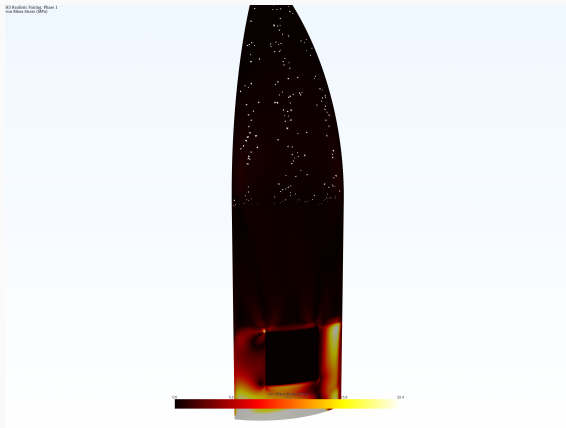
Quantities of interest:

- Peak von Mises stress (limit 1,200 MPa)
- Max cohesive damage SDEG (limit 0.5)
- Max displacement



Non-intrusive: only input files and output databases are touched — no solver source is modified.

Abaqus FE solve — von Mises stress field



One deterministic solve at one quadrature node.

- Each PCE quadrature node is one **deterministic Abaqus solve**.
- Real **CFRP / Al-honeycomb** sandwich fairing (Abaqus ODB); von Mises peaks around the **access-door cutout**.
- The PCE study wraps a representative **sandwich panel** of this structure as a black box.

Result 1: response statistics converge at degree 2

QoI	Mean	Std. Dev.	CV	P_f	β
$\sigma_{vM}^{\max}$ (deg 2)	209.7 MPa	2.375 MPa	1.13%	0	∞
$\sigma_{vM}^{\max}$ (deg 3)	209.7 MPa	2.378 MPa	1.13%	0	∞
SDEG	0.000	0.000	—	0	∞
$u^{\max}$	10.07 mm	0.178 mm	1.76%	—	—

Degree 2 (66 runs) and degree 3 (286 runs) agree to **four significant figures**.

This is more than a convergence check — it *quantifies the curvature* of the response surface. Had higher-order terms mattered, the extra 220 simulations would have moved the standard deviation. They did not.

Takeaway

Degree 2 is the **correct model order**, not an economy measure.

SDEG $\equiv$ 0 in all **352** runs.

Result 2: E_1 governs over 99% of the variance

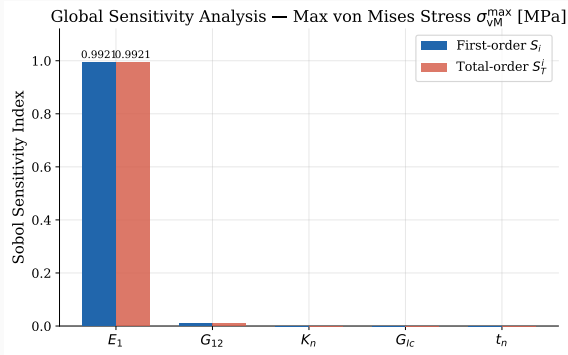
	S_i stress	S_i disp.
E_1	0.9921	0.9995
G_{12}	0.0079	0.0005
K_n, G_{Ic}, t_n	≈ 0	≈ 0

$S_i \approx S_T^i$: inputs act additively, no resolvable interactions.

What this means

The 5-D problem collapses to **effectively 1-D** in E_1 .

G_{12} carries *twice* the CoV of E_1 yet contributes < 1% — sensitivity, not scatter, sets the ranking.



S_i (blue) beside S_T^i (red). The pairs coincide, so no variance is held in interactions; the cohesive bars vanish because damage never initiates.

Why does E_1 dominate? A first-order argument

$$\text{Var}(Y) \approx \sum_{i=1}^d \left(\frac{\partial Y}{\partial \xi_i} \right)^2 \sigma_{\xi_i}^2$$

- Each contribution scales with **sensitivity** $\times$ **variance** — not either alone.
- Panel carries load by **membrane action**: stress and deflection set by axial stiffness $E_1 t \Rightarrow$ large $\partial Y / \partial E_1$, and E_1 still carries 5% scatter.
- G_{12} has *twice* the CoV but tiny $\partial Y / \partial G_{12} \Rightarrow$ suppressed.

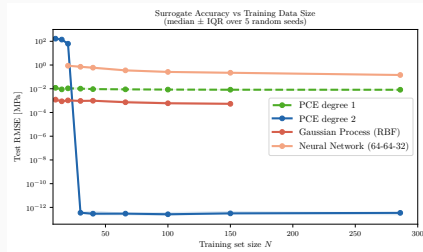
The rule

Ranking is set by which uncertainty the structure is configured to *amplify* — not by which input is most uncertain.

Result 3: PCE vs neural-network surrogate

	LOO RMSE	PCE deg 2	PCE deg 3	NN (66)	NN (286)
$\sigma_{vM}^{\max}$ (MPa)		9.4×10^{-4}	1.5×10^{-3}	7.2×10^{-4}	1.8×10^{-2}
$u^{\max}$ (mm)		4.2×10^{-5}	7.1×10^{-5}	2.7×10^{-5}	1.6×10^{-3}

- At $N = 66$ both are accurate to $\sim 10^{-3}$ MPa — the NN marginally lower on this metric.
- The difference is **stability**: NN error **grows** $25\times$ from 66 to 286 points, while PCE deg. 2 holds; deg. 3 is slightly *worse* (over-parameterisation).
- PCE returns Sobol indices analytically; the NN would need a separate Monte Carlo study.

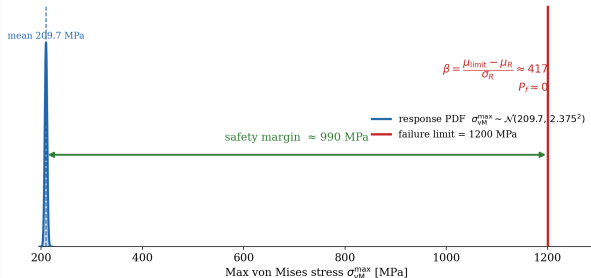


Test RMSE vs training-set size, median and IQR over 5 seeds. The

NN curve turns *upward* — more data, worse fit.

Reliability: a wide margin, and an unidentifiable interface

Reliability: the response sits far below the failure limit



A degree-2 PCE surrogate places the entire stress distribution far below the limit; cohesive damage never initiates (SDEG = 0).

Density and limit on one axis. The PDF is the narrow spike on the left: $\pm 5\sigma$ spans ≈ 24 MPa against a 990 MPa margin. No tail shape survives that separation — hence $P_f = 0$ exactly.

$$P_f = P(\sigma_{vM}^{\max} > 1200) = 0, \quad \beta = \infty$$

- Mean stress is $(1200 - 209.7)/2.38 \approx 416\sigma$ below failure.
- Read $P_f = 0$ as “**below the resolution** of the analysis” (MC of 10^6 ; inputs truncated at $\pm 50\%$).
- $SDEG \equiv 0 \Rightarrow$ cohesive parameters are **structurally unidentifiable** at this load.

Extension T9: non-Gaussian KL random field for E_1

KL expansion of an anisotropic exponential kernel, then:

- Gaussian field admits **negative stiffness** $\Rightarrow$ **translation field** $H = F^{-1}(\Phi(Z))$: lognormal, positive by construction.

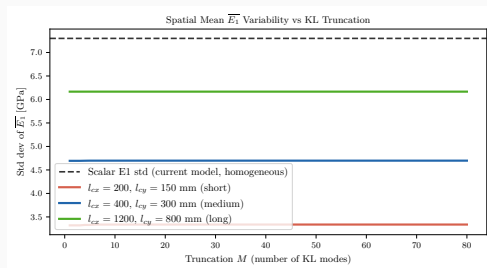
- Preserves rank, *not* linear correlation:

$$\rho_H = \frac{e^{\sigma_{\ln}^2 \rho_Z} - 1}{e^{\sigma_{\ln}^2} - 1} \text{ — distortion } 3 \times 10^{-4} \text{ (CoV 5\%)} \text{ to } 3.8 \times 10^{-2} \text{ (60\%)}.$$

- Spatial averaging: $\sigma(\overline{E_1}) \approx 4,700 < 7,300$ MPa.

Scope of the claim

Conservative for **average-governed** responses — not for peak stress.



Std of $\overline{E_1}$ vs KL truncation M , three correlation-length pairs; dashed = scalar model, 7.3 GPa. Longer correlation $\Rightarrow$ less averaging $\Rightarrow$ closer to the scalar case.

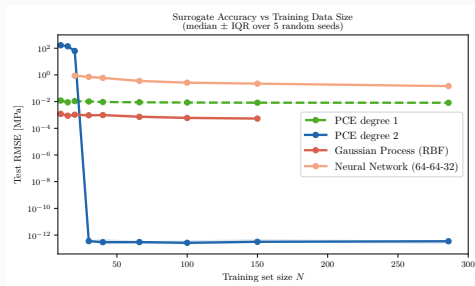
Extension T13: a correct prior beats model flexibility

PCE (deg. 1/2), **GP** (RBF), **NN** (MLP) over $N = 10\text{--}286$. Median RMSE, 5 seeds.

Surrogate	RMSE at $N=66$
PCE deg. 2	3.0×10^{-13} MPa
GP	7.3×10^{-4} MPa
NN	3.6×10^{-1} MPa

Read this correctly

Reference response *is* a deg.-2 polynomial $\Rightarrow 10^{-13}$ measures **exact representability**, not generalisation. The meaningful factor ($\approx 500\times$) is **GP vs NN**.



Test RMSE vs training-set size, median and IQR over 5 seeds. Only the GP carries an error bar of its own: epistemic std 0.008 MPa against an aleatory scatter of 2.37 MPa = **0.3%**, so surrogate error is negligible beside the physical variability.

Extension T6: E_1 dominance $\Rightarrow$ tractable Bayesian identification

PCE linear term: $\hat{\sigma}_{\text{vM}} = 209.7 + 2.365 \xi_{E_1}$ MPa.

Conjugate Gaussian update on the mean of k measurements:

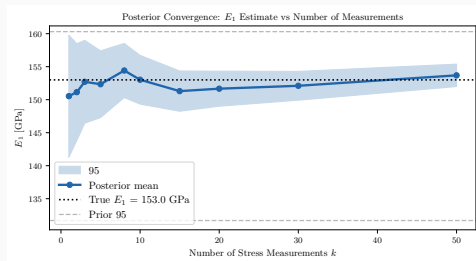
$$\sigma_{\text{post}}(E_1) = \frac{\sigma_{E_1}}{\sqrt{1 + k c_1^2 / \sigma_{\text{meas}}^2}}$$

Key result: posterior std falls from 7,300 MPa to 2,582 ($k=5$), **1,885** ($k=10$, -74%), 1,356 ($k=20$).

What it does and does not do

Reduces **epistemic** uncertainty $\Rightarrow$ tolerance allocation.

P_f is already 0 at 416σ , so reliability is *unchanged*.



Posterior mean of E_1 and its 95% credible band against the number of measurements k . Each k uses the mean of k *independent* draws, so the band contracts as $1/\sqrt{k}$ and stays centred on the true value (dotted). Dashed grey is the prior 95% interval — the width the data has to beat.

Conclusions & limitations

- PCE deg. 2 (66 runs) converged; E_1 **controls** > 99% of variability; panel reliable at 416σ .
- **T9**: lognormal translation field guarantees positivity; spatial averaging cuts $\sigma(\overline{E_1})$ to 4,700 MPa — conservative for *average-governed* responses.
- **T13**: a correct prior beats flexibility at small N ; the 10^{-13} is exact representability, the meaningful gap (500 $\times$) is GP vs NN.
- **T6**: 10 measurements $\Rightarrow -74\%$ posterior std — **epistemic** reduction, for tolerance allocation, not for reliability.

Honest limitations

$P_f = 0$ is resolution-limited; inputs assumed independent; single panel, not full fairing; cohesive parameters unidentifiable at this load; KL realisations never propagated through the solver. **Next**: activate cohesive zone; correlated input model; map the field into the FE run.